

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist who turns complex physical models into fast, tested software and turns large scientific datasets into calibrated models. Ph.D. in Physics and ten years building simulation codes and the data pipelines around them: author of Paramo, an open-source Fortran/OpenMP radiative-transfer library used by other research groups; led a 7-person upgrade of the PatchworkMHD black-hole simulation code on the Frontera supercomputer; mapped wetland probability across a 2,160 km² watershed with random-forest models. Nine peer-reviewed articles and Co-I on a funded NASA Fermi Guest Investigator grant.

Skills

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, random forest, SVM, neural networks, gradient-based optimization, parameter estimation

Scientific Computing & HPC: Numerical PDEs (Fokker–Planck, advection–diffusion), radiative transfer, finite-difference/finite-volume methods, OpenMP, MPI, OpenACC, SLURM, PBS, HDF5

Geospatial & Remote Sensing: Sentinel-1/2 multispectral imagery, DEM terrain analysis, raster/vector pipelines, QGIS, geospatial feature engineering

Languages: Python, C/C++, Fortran, Rust, R, SQL, Java, Julia, Shell

Tools: Git, Docker, Azure, Jenkins, OpenShift, Kafka, Splunk, Jira, ParaView, VisIt

Experience

Independent Contractor

Jan 2026 – Present

Mercor

Remote

Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models and co-developed others with fellow writers.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.

Spatial Data Scientist

May 2025 – Jan 2026

TealWaters

Seattle, WA

Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models.
- Trained 2 colleagues on the WIP tool and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer

Jan 2024 – May 2025

Self-directed

Seattle, WA

Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker–Planck equation coupled to radiative transfer; published in ApJ (2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.

Software Engineer

Apr 2022 – Jan 2024

Paychex

Remote

Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts that flagged production incidents.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.
- Prepared, validated, and analyzed Oracle EBS data with SQL.

Postdoctoral Research Associate

Rochester Institute of Technology

Feb 2021 – Apr 2022

Rochester, NY

HPC simulations of supermassive black-hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black-hole binaries.
- Added black-hole spin to the simulations with under 5% runtime overhead.
- Benchmarked new features on TACC's Frontera supercomputer (about 1,200 node-hours) to find performance bottlenecks.
- Calibrated Paramo against Fermi-LAT spectra with gradient-based optimization; supervised a graduate student on the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron-star mergers.

Postdoctoral Research Fellow

Purdue University

Oct 2018 – Nov 2020

West Lafayette, IN

Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Architected Paramo, an open-source Fortran 95 Fokker–Planck and radiative-transfer library; parallelized and optimized it with OpenMP to run about $20\times$ faster than serial on 5 cores.
- Paramo has been adopted by other research groups, including in Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded (\$68K) to model blazar radiation.
- Mentored 3 graduate students, co-authoring papers with each.

Postdoctoral Research Fellow

Universidad Michoacana de San Nicolás de Hidalgo

Jan 2018 – Sep 2018

Morelia, Mexico

- Built a Python pipeline that turned GRTrans black-hole simulation images into training data for an SVM classifier of black-hole spin.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant

Universitat de València

Oct 2011 – Jul 2017

Valencia, Spain

- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape.
- Published 2 first-author papers in MNRAS.

Projects

Blazar Population ML

Scientific ML (Python, scikit-learn, PyTorch)

Jan 2026 – Present

github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Building a neural network that maps physical jet parameters to observable spectral energy distributions accurately enough to compare with Tleco results.
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

Selected Publications (9 peer-reviewed articles, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

Education

Universitat de València

Ph.D. in Physics, Excellent Cum Laude

Valencia, Spain

2017

Universidad Michoacana de San Nicolás de Hidalgo

M.Sc. in Physics

Morelia, Mexico

Universidad Autónoma del Estado de México

B.Sc. in Physics

Toluca, Mexico